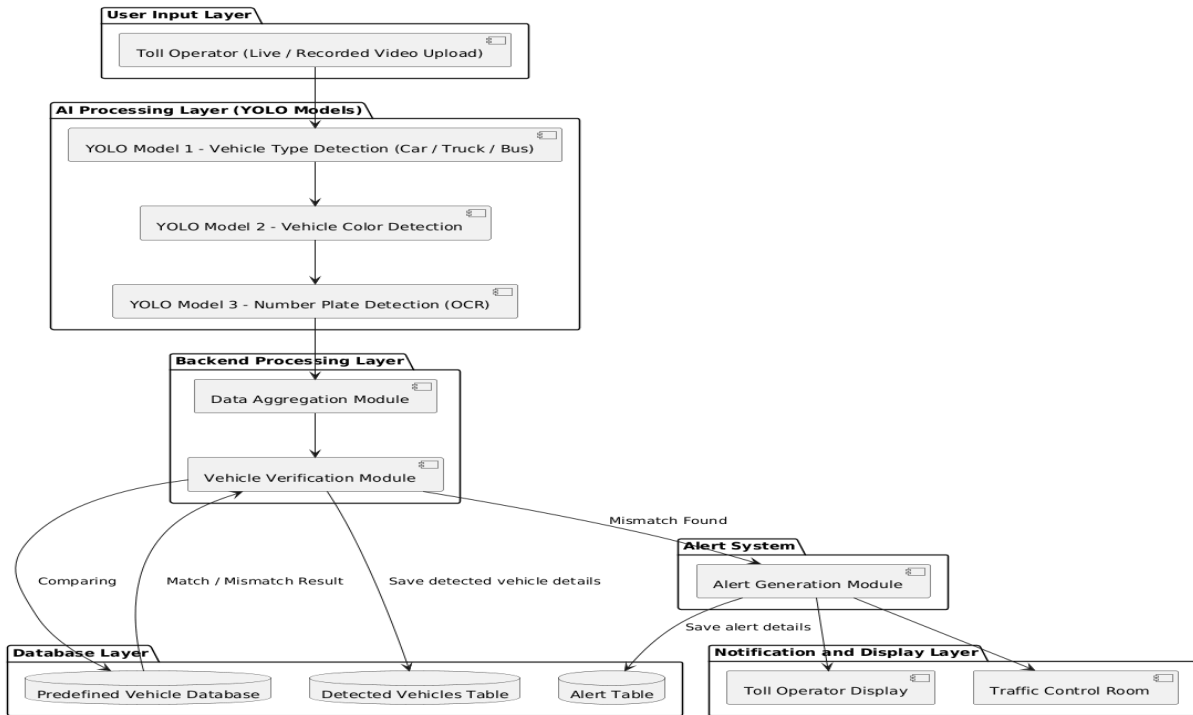
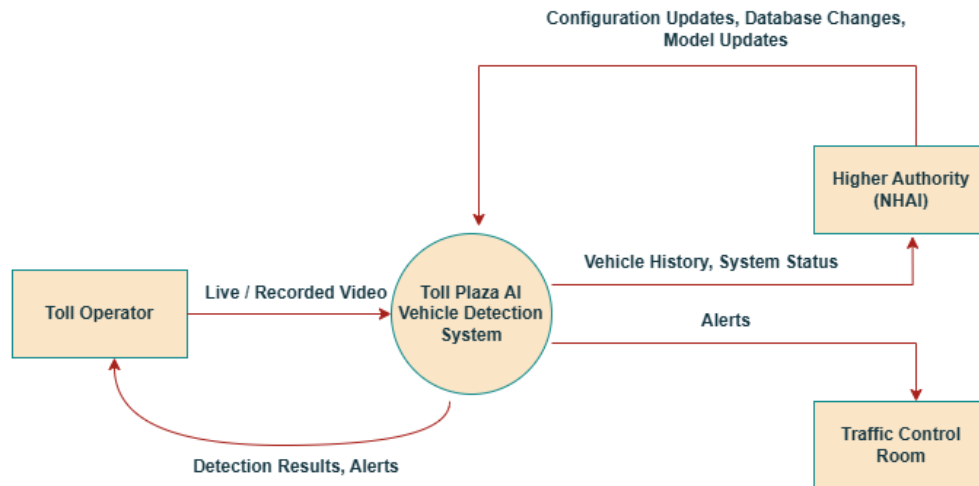


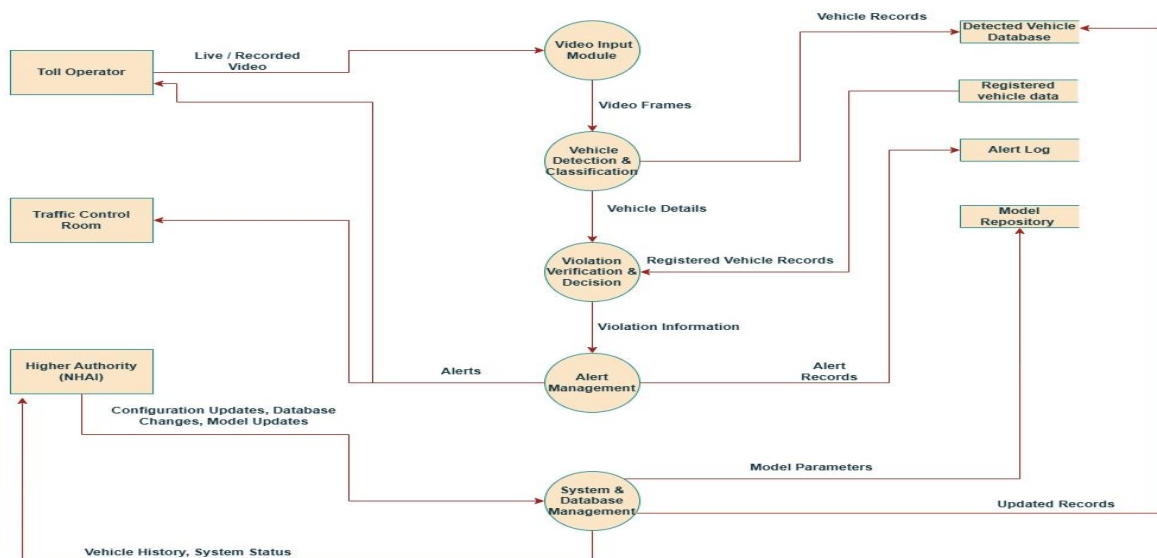
System Architecture



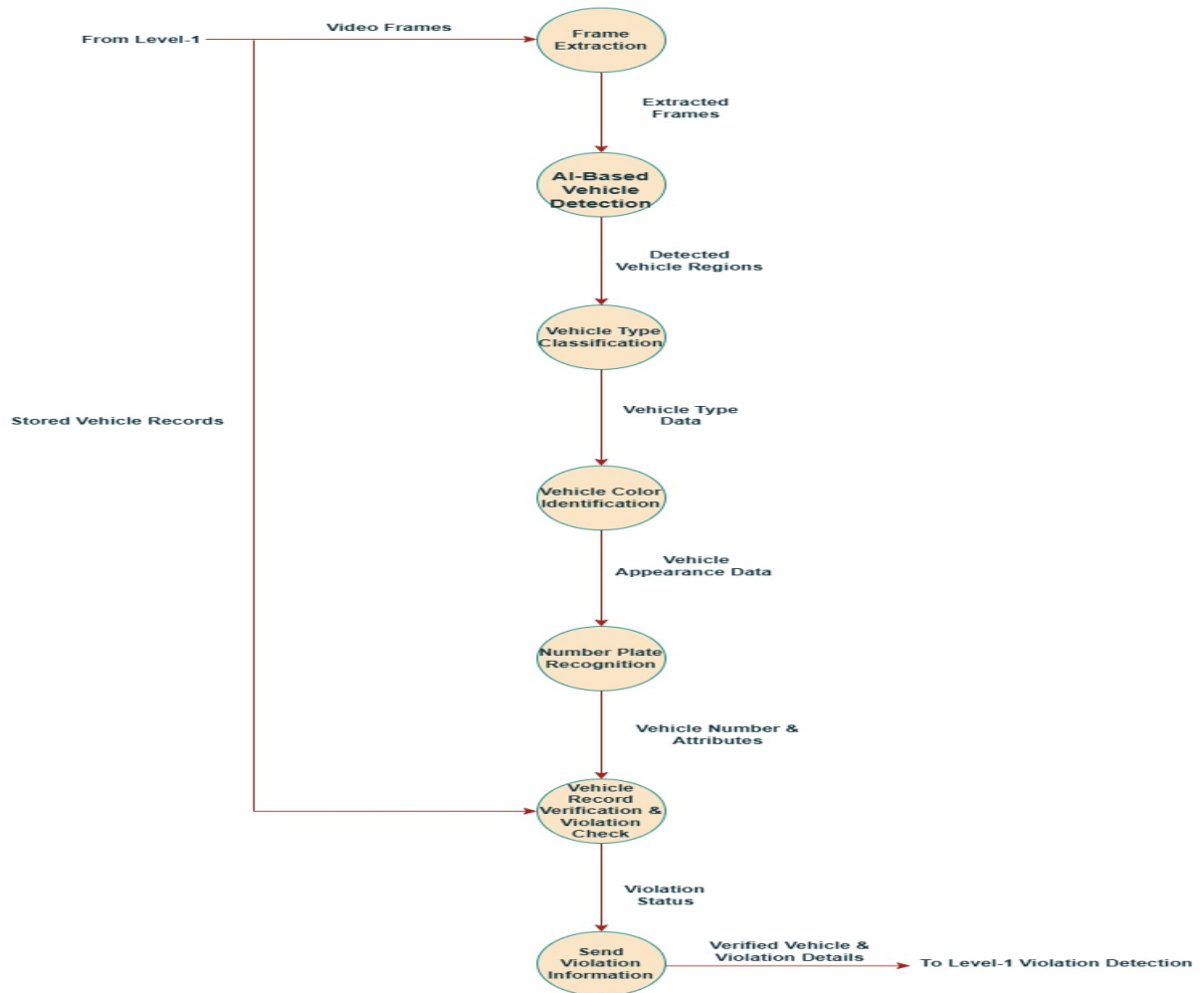
Data Flow Diagram



DFD Level 0

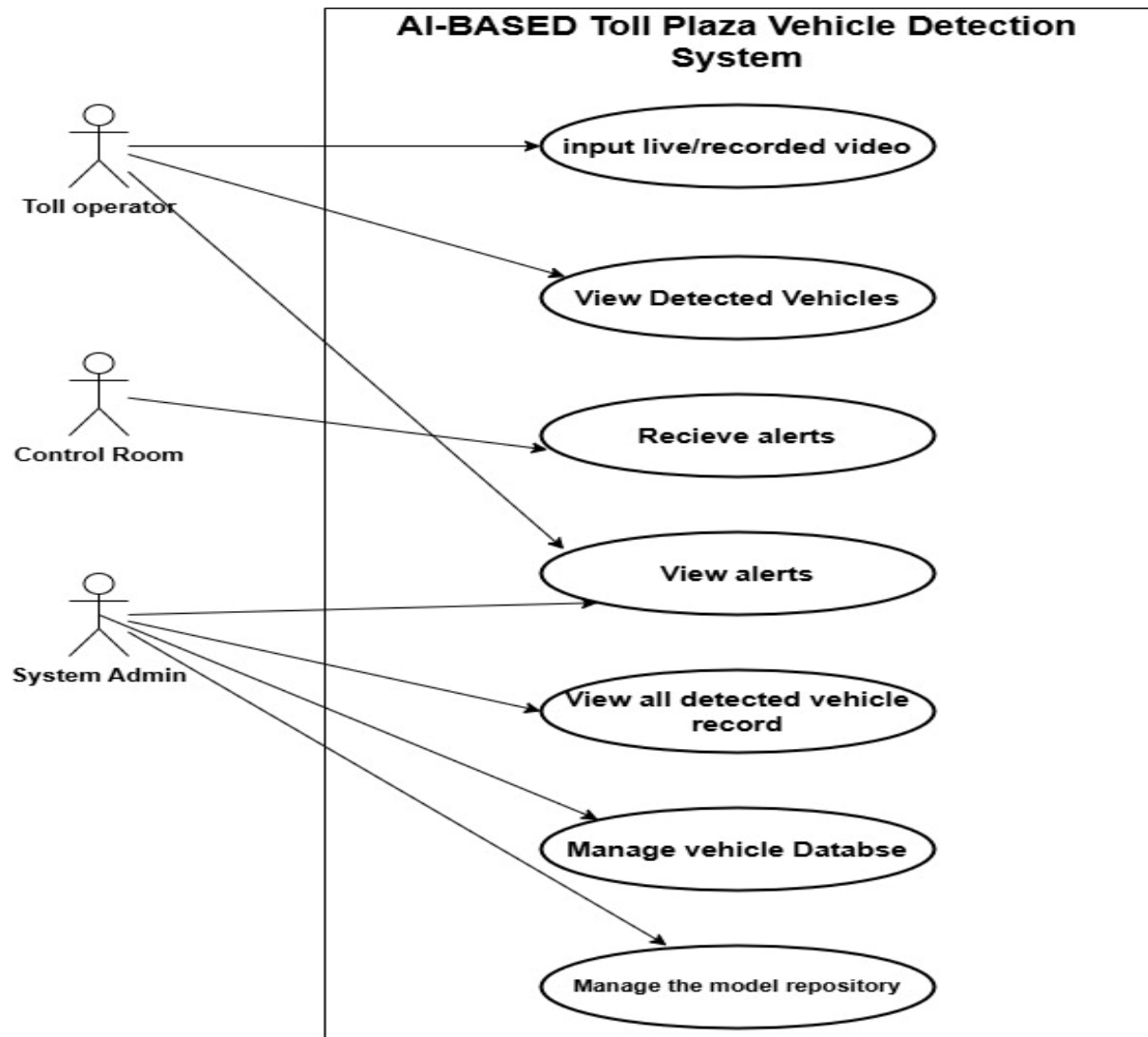


DFD LEVEL 1



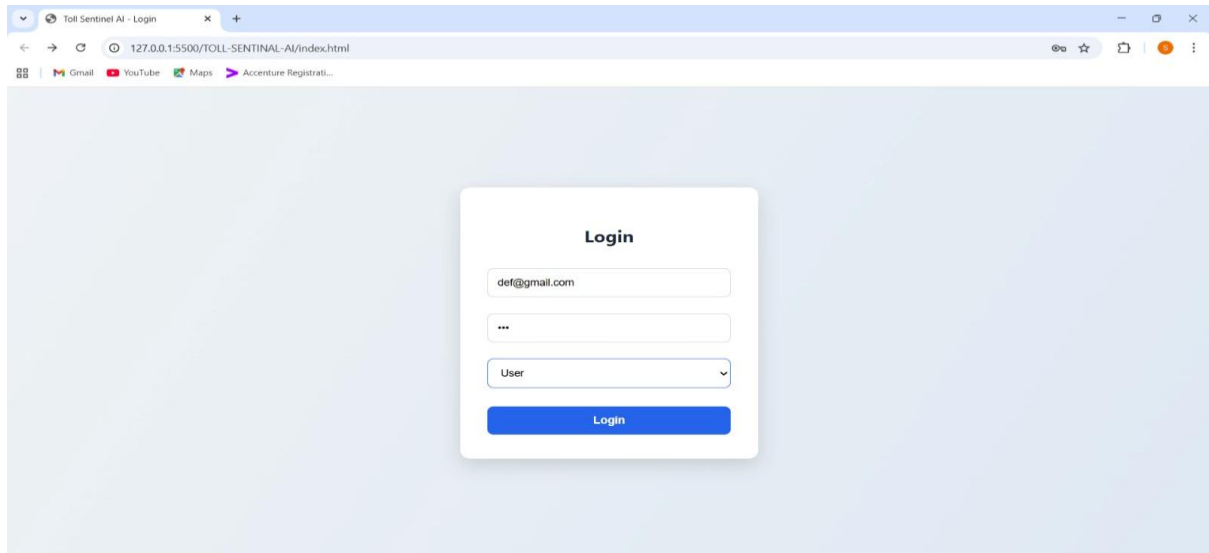
DFD Level-2 (Expansion of Vehicle Detection & Classification – Level-1 Process)

Use Case Diagram



GUI Design

Toll operator



A screenshot of a web browser showing the login page for 'Toll Sentinel AI'. The browser's address bar displays '127.0.0.1:5500/TOLL-SENTINEL-AI/index.html'. The login form is centered on a light blue background. It features a title 'Login', an email input field with 'def@gmail.com', a password input field with three dots, a dropdown menu currently showing 'User', and a blue 'Login' button.

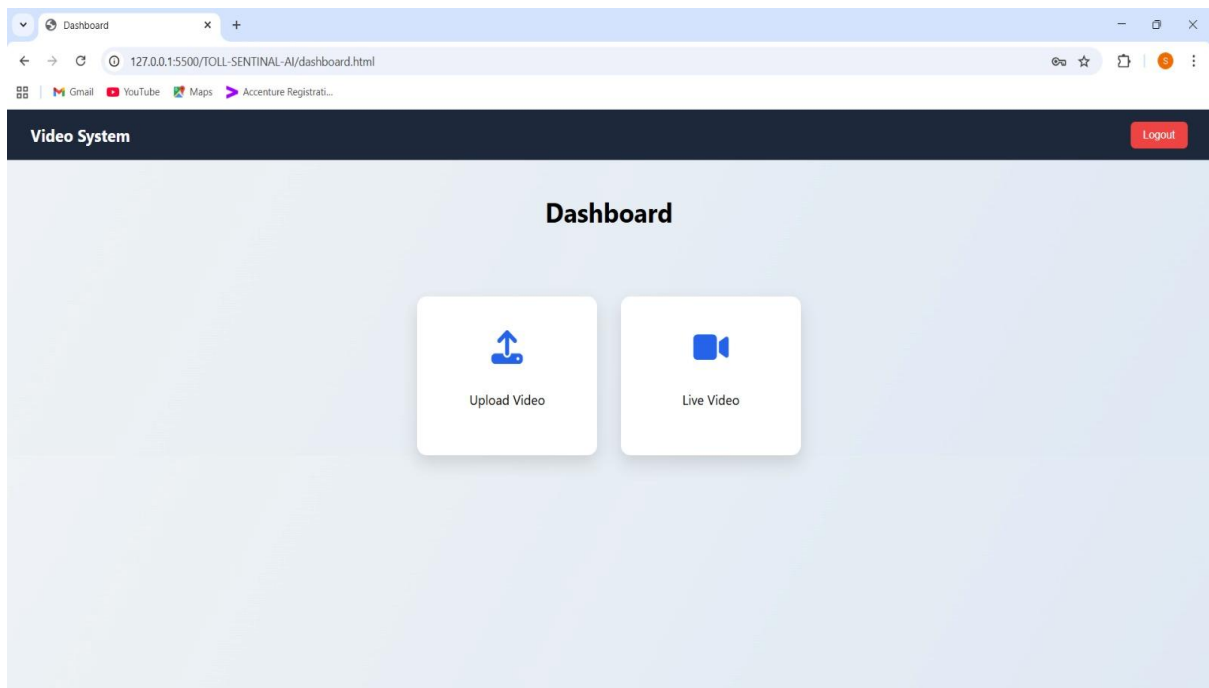
Login

def@gmail.com

...

User

Login



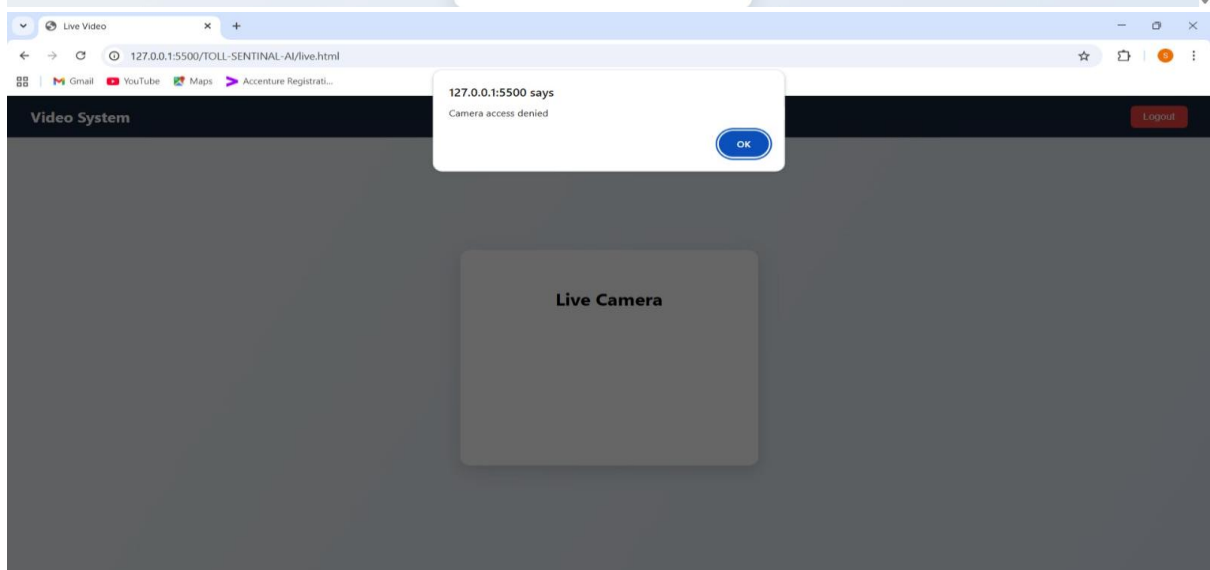
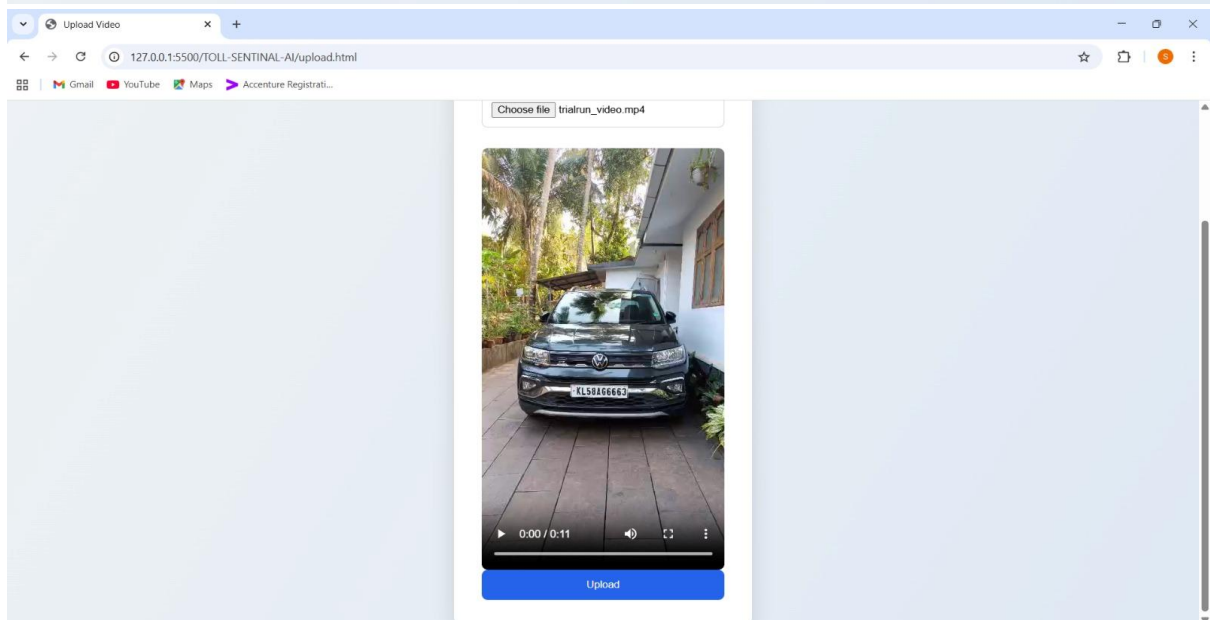
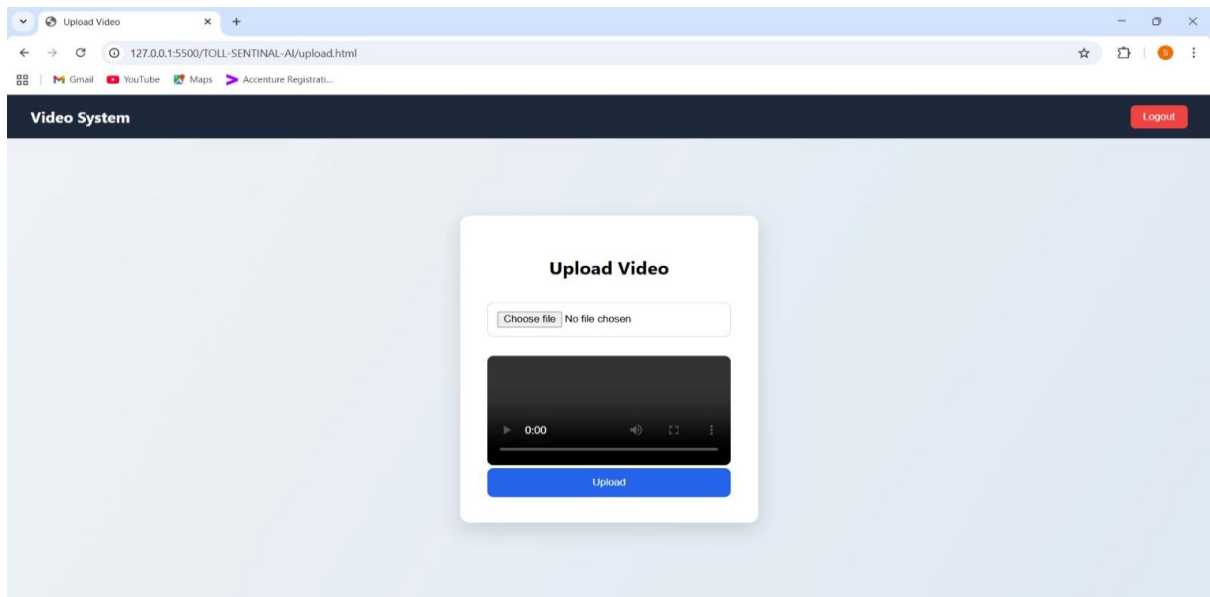
A screenshot of the 'Dashboard' page for the 'Video System'. The browser's address bar shows '127.0.0.1:5500/TOLL-SENTINEL-AI/dashboard.html'. A dark header bar at the top contains the text 'Video System' and a red 'Logout' button. The main content area has a title 'Dashboard' and two white buttons with blue icons: 'Upload Video' (with an upload icon) and 'Live Video' (with a video camera icon).

Video System Logout

Dashboard

Upload Video

Live Video



Admin

Toll Sentinel AI - Login

127.0.0.1:5500/TOLL-SENTINAL-AI/index.html

Gmail YouTube Maps Accenture Registrati...

Login

abc@gmail.com

Admin

Login

Admin Dashboard

127.0.0.1:5500/TOLL-SENTINAL-AI/admin_dashboard.html

Gmail YouTube Maps Accenture Registrati...

Higher Authority Dashboard

Logout

All Vehicle Details

Edit

Vehicle No	Color	Type	Status	Owner Name	Contact Info	Action
KL-07-AB-1234	White	Car	Verified	Rahul	9876543210	Delete
KL-10-CD-5678	Black	Truck	Pending	Anil Kumar	9123456789	Delete

Add New Vehicle

Detected Vehicles

Detection records are updated every 6 months.

Detection ID	Vehicle Reg No	Detected Colour	Detected Type	Detection Time	Lane No	Toll ID
101	KL-07-AB-1234	White	Car	2026-01-15 10:32	2	TOLL-01

Alerts

Admin Dashboard

127.0.0.1:5500/TOLL-SENTINAL-AI/admin_dashboard.html

Gmail YouTube Maps Accenture Registrati...

101	KL-07-AB-1234	White	Car	2026-01-15 10:32	2	TOLL-01
102	KL-10-CD-5678	Black	Truck	2026-01-18 14:05	1	TOLL-02

Alert_Table

Alert ID	Vehicle Reg No	Alert Reason	Alert Time	Toll ID	Toll Location	Owner Name	Owner Contact No
A201	KL-10-EF-9999	Invalid Toll Tag	2026-02-02 09:45	TOLL-03	Ernakulam	Arjun Kumar	9876543210
A202	KL-08-GH-4321	Blacklisted Vehicle	2026-02-03 16:20	TOLL-01	Thrissur	Suresh Nair	9123456789

Add New User

User Name

Email

Add User

Database Design

Registered_vehicle_data table

Field Name	Data Type	Description
Vehicle Reg.No	VARCHAR(PK)	Vehicle registration number
Vehicle colour	VARCHAR	Registered vehicle colour
Vehicle type	VARCHAR	Car / Truck / Bus / Bike
Vehicle Status	VARCHAR	Normal / Stolen / Blacklisted
Owner Name	VARCHAR	Vehicle owner name
Owner contact no	VARCHAR	Owner contact number

Detected_Vehicle_Table

Field Name	Data Type	Description
Detection_id	INT(PK)	Detection record ID
Vehicle_reg_no	VARCHAR	Detected vehicle number
Detected_colour	VARCHAR	Colour detected by AI
Detected_type	VARCHAR	Vehicle type detected
Detection_time	DATETIME	Date and time of detection
Lane_no	INT	Lane number
Toll_id	INT	Toll plaza ID

Alert_Table

Field Name	Data Type	Description
Alert_id	INT (PK)	Unique alert ID
Vehicle_reg_no	VARCHAR	Vehicle involved
Alert_reason	VARCHAR	Stolen / Blacklisted / Mismatch
Alert_time	DATETIME	Time of alert generation
Toll_id	INT	Toll plaza ID
Toll_location	VARCHAR	Physical location of the toll plaza
Owner_name	VARCHAR	Owner name
Owner_contact_no	VARCHAR	Owner contact number